

Econ 2 - Lecture 5 - 4/13/26

Midterm Exam in 2 weeks → 30 Multiple Choice ?'s
Chapters 1, 2, 3 → Lots of Resources, create more
Discussion Activity #2 this week

Last Class: GDP calculation

<u>Product</u>	2009		2016	
	<u>Quantity</u>	<u>Price</u>	<u>Q</u>	<u>P</u>
Food	1000	1	1500	2
Housing	100	100	150	150
Movies	500	5	1000	10

Nominal GDP: Value of transactions,
actual dollars exchanged

$$\text{Nom. GDP}_{2009} = Q_{09}^F \times P_{09}^F + Q_{09}^H \times P_{09}^H + Q_{09}^M \times P_{09}^M \\ = 1000 \times 1 + 100 \times 100 + 500 \times 5 = 13,500$$

$$\text{Nom. GDP}_{2016} = Q_{16} \times P_{16} \text{ (Multiply \& add)} \\ = 1500 \times 2 + 150 \times 150 + 1000 \times 10 \\ = 35,500$$

Nominal GDP increased by ~3x, why?

Price ↑, Quantity ↑

Not more production → need to separate price changes
from quantity changes

Real GDP: Calculate GDP using the same base year prices (fixed) every year

Step 1: Define a base year, any year we choose
→ Cannot change after fixed ⇒ Base Year = 2009
BY = 2009

Step 2: Calculate GDP using base year prices

$$\text{Real GDP}_{2016} = Q_{16}^F \times P_{BY=2009}^F + Q_{16}^H \times P_{BY}^H + Q_{16}^M \times P_{BY}^M$$

$$= 1500 \times 1 + 150 \times 100 + 1000 \times 5$$

$$\text{Real GDP}_{2016} = 21,500$$

(BY = 2009)

If prices did not change between 2009 & 2016

the nominal GDP in 2016 would have been 21,500
→ not 35,500 ⇒ includes price changes

Note: Real GDP = $Q_{CY} \times \underline{P_{BY}}$ (Multiply & Add)

current year (CY) fixed for every year

What is Real GDP in Base Year? Set CY = BY

$$\text{Real GDP}_{CY=2009} = Q_{2009} \times P_{BY=2009} \Rightarrow \text{Nominal GDP}$$

Separate price increase from quantity increase

GDP Deflator $\Rightarrow$ Weighted Price Ratio

$$\begin{aligned}\text{GDP Def}_{cy} &= \frac{\text{Nominal GDP}_{cy}}{\text{Real GDP}_{cy}} \times 100 \\ &= \frac{Q_{cy} \times P_{cy}}{Q_{cy} \times P_{by}} \times 100\end{aligned}$$

$$\text{GDP Deflator}_{cy=by=2009} = \frac{Q_{cy=2009} \times P_{cy=2009}}{Q_{cy=2009} \times P_{by=2009}} \times 100 = 100$$

Base Year GDP Deflator = 100

$$\text{GDP Def}_{2016} = \frac{35,500}{21,500} \times 100 = 165$$

If average price of a good in base year was 100 in current year (2016), the average price increased to 165

Prices increased 65% between 2009 & 2016

Convert GDP Deflator to Inflation Rate

$$\begin{aligned}\text{Percent Change in GDP Deflator}_{y_1, y_2} &= \frac{\text{GDP Def}_{y_2} - \text{GDP Def}_{y_1}}{\text{GDP Def}_{y_1}} \times 100\end{aligned}$$

$$\text{Inflation Rate}_{y_1, y_2} = \frac{165 - 100}{100} \times 100 = \frac{65}{100} \times 100 = 65\%$$

$$\text{GDP Deflator}_{2017} = 172$$

$$\text{Inflation Rate}_{16,17} = \frac{\text{GDP Def}_{17} - \text{GDP Def}_{16}}{\text{GDP Def}_{16}} \times 100$$

$$= \frac{172 - 165}{165} \times 100 = \frac{7}{165} \times 100 = 4.24\%$$

Separate Quantity Increases

$$\text{Growth Rate in Real GDP}_{2007,2016} = \frac{\text{Real GDP}_{2016} - \text{Real GDP}_{2007}}{\text{Real GDP}_{2007}} \times 100$$

$$= \frac{21,500 - 13,500}{13,500} \times 100$$

$$\text{Real GDP Growth} = 59\%$$

Shortcomings of GDP Statistic

Informed labor market / production

Quality Adjustments

Old Laptop Price = \$1000
New Laptop (today) = \$1200

Quality adjust laptop $\approx 1200 - 350$
 $= \$850$

Bureau of Econ.
Analysis

Estimate value of
additional computing
power $\rightarrow$ Example \$350
worth of additional power

Goal #1: High Standard of Living
→ GDP: Strength & Limitations

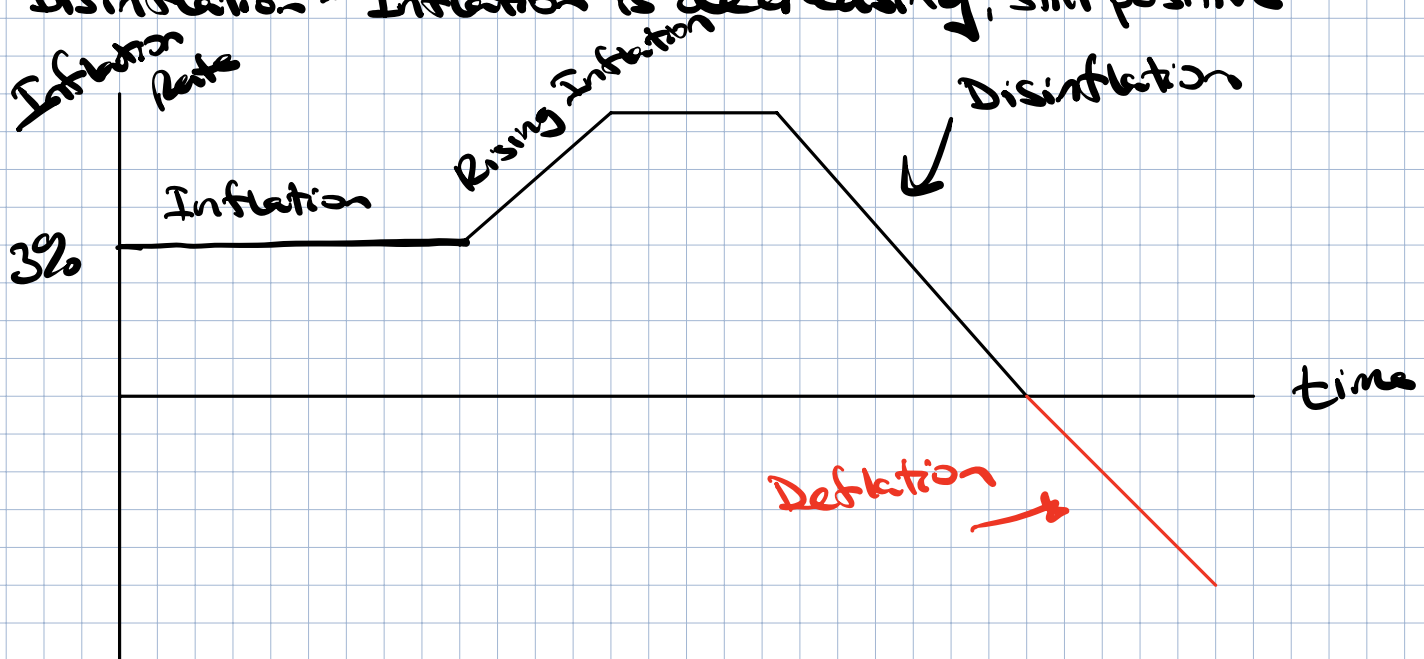
Goal #2: Stable Prices

GDP Deflator: measure of prices using every single good/service in GDP

Inflation = Prices are rising

Deflation = Negative inflation = Prices are falling

Disinflation = Inflation is decreasing, still positive



Stable Price Goal ⇒ Federal Reserve

Inflation Rate = 2% ⇒ Low and predictable
Target inflation rate

↙
Anchored
at 2%

If target = 100% ⇒ Double every year!

A small "miss" ⇒ big problems ⇒ affordability

Missing a low target ⇒ less costly

Why not target 0%? Hard to hit exactly,

Risk Deflation $\rightarrow$ Deflationary Spiral $\rightarrow$ target above 0%

Prices are expected to fall $\rightarrow$ Postpone buying

↑
fewer jobs ← Less Production
↓

GDP Deflator $\rightarrow$ inflation $\rightarrow$ all GDP production!

Do consumers care about the price of every good produced?

Create a new measure of inflation

Consumer Price Index (CPI)

Only consider prices of goods relevant to consumers

Build CPI $\rightarrow$

Step 1: Create a basket of goods,
households care about

$\rightarrow$ food, Housing, gas, coffee, education, alcohol,
cars, subscriptions, entertainment, airline tickets, etc
Consumption (C)

$\rightarrow$ Used goods? Cars, homes, etc $\rightarrow$ include (not in GDP)

$\rightarrow$ Import Goods: Wine from France, bought in CA (include)

$\rightarrow$ Exported Goods: Wine from CA, bought in France (exclude)

Include Cons. (C) & Imports (M)

Exclude Investment (I), Gov't Spending (G) &
Exports (X)

Step 2: Weight the importance of every
good in basket relative to budget